Meet the Lab Worksheet

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Course/Section 002

Please complete the following steps to complete the introduction lab:

1. What color wire carries 5V on the breadboard? Red wire
2. What color distribution strip carries 5V on the breadboard? Red strip
3. What color wire carries 0V on the breadboard? Black wire
4. What color distribution strip carries 0V on the breadboard? Blue strip
5. Using a type A to micro B USB cable, connect the breadboard to the USB port of the computer.
6. Using a multi-meter, measure the difference between the red and black wires on the board and record below. It should be around 5V, but it will not be exactly.

   The voltage was 5.047 V

Note: This is the same voltage that is provided your flash drives and phone chargers when you plug them into a computer.

7. Take the blue LED and a 220 ohm resistor (red-red-brown) in series as shown in the LED Section above and connect them across the red (5V) and blue (0V) distribution strips at a point near the Digilent label so it is out of the way for the next steps. It should glow brightly.

8. Take the orange LED and a 220 ohm resistor (red-red-brown) in series as shown in the LED Section above and connect them between the center connection of one of the slide switches and the blue (0V) distribution strips.

9. Using a multi-meter, measure the difference between the center connection of the switch and the blue (0V) distribution strip and record below with the switch in the left and right position below:

With the switch in the left position (relative the the picture in the Breadboard Section),
the voltage was 5.032 V and the LED was Off (on or off)

With the switch in the right position,
the voltage was 4.08 V and the LED was On (on or off)

10. Lab checkout steps: